COMP9020 Week 9
Term 3, 2020
Combinatorics

- [LLM] - Ch. 14
- [RW] - Ch. 5, 7
- [Rosen] - Ch. 6, 8

Counting Techniques

General idea: find methods, algorithms or precise formulae to count the number of elements in various sets or collections derived, in a structured way, from some basic sets.

Examples

Single base set $S = \{s_1, \ldots, s_n\}$, $|S| = n$; find the number of

- all subsets of S
- ordered selections of r different elements of S
- unordered selections of r different elements of S
- selections of r elements from S such that ...
- functions $S \longrightarrow S$ (onto, 1-1)
- partitions of S into k equivalence classes
- graphs/trees with elements of S as labelled vertices/leaves

Example

Example

A restaurant has the following menu:

Starter	Main Course	Dessert
Soup	Fish	Ice-cream
Bread	Beef	Fruit
	Pork	Cheese
	Chicken	

How many:

- 3 course meals (Starter-Main-Dessert) are possible?
- 3 course meals (Any item for each course) are possible?
- 3 course meals (Any item, no duplicates) are possible?
- Meals consisting of 3 items (order is unimportant)?

Applications of counting in CS

- Algorithmic analysis
- Data management
- Enumeration techniques
- Probability calculations

Example

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How many:

- Starter-Main-Dessert? $2 \times 4 \times 3 = 24$
- Any item for 3 courses?
- Any item, no duplicates, for 3 courses?
- Meals of 3 items?

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- Any item for 3 courses? $9 \times 9 \times 9 = 729$
- Any item, no duplicates, for 3 courses?
- Meals of 3 items?

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- Any item, no duplicates, for 3 courses? $9 \times 8 \times 7 = 504$
- Meals of 3 items?

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- Any item for 3 courses? $9 \times 9 \times 9 = 729$
- Any item, no duplicates, for 3 courses? $9 \times 8 \times 7 = 504$
- Meals of 3 items? $504/6 = 84$

Outline

- Basic counting rules
- Combinations and Permutations
- Alternative techniques
- Difficult counting problems

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Basic Counting Rules: Principles

Two simple rules:

- **Union rule** ("or"): If S and T are disjoint $|S \cup T| = |S| + |T|$
- **Product rule** ("followed by"): $|S \times T| = |S| \cdot |T|$

These cover many examples, though the rule application is not always obvious.

Common strategies:

- Direct application of the rule
- Relate unknown quantities to known quantities (e.g. $|S| + |T| = |S \cup T| + |S \cap T|$)
- Find a bijection to a set that can be counted

Basic Counting Rules (1)

Union rule — S and T *disjoint*

$$|S \cup T| = |S| + |T|$$

$S_1, S_2, \ldots, S_n$ pairwise disjoint ($S_i \cap S_j = \emptyset$ for $i \neq j$)

$$|S_1 \cup \ldots \cup S_n| = \sum |S_i|$$

Example

How many numbers in $A = [1, 2, \ldots, 999]$ are divisible by 31 or 41?

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$$|S_1 \cup \ldots \cup S_n| = \sum |S_i|$$

Example

How many numbers in $A = [1, 2, \ldots, 999]$ are divisible by 31 or 41?

$\lfloor 999/31 \rfloor = 32$ divisible by 31
$\lfloor 999/41 \rfloor = 24$ divisible by 41
No number in A divisible by both
Hence, $32 + 24 = 56$ divisible by 31 or 41

Basic Counting Rules (1)

Union rule: Inferences

For arbitrary sets S, T, ...

$$
\begin{aligned}
|S \cup T| &= |S| + |T| - |S \cap T| \\
|T \setminus S| &= |T| - |S \cap T| \\
|S_1 \cup S_2 \cup S_3| &= |S_1| + |S_2| + |S_3| \\
&\quad - |S_1 \cap S_2| - |S_1 \cap S_3| - |S_2 \cap S_3| \\
&\quad + |S_1 \cap S_2 \cap S_3|
\end{aligned}
$$

Basic Counting Rules (2)

Product rule

$$|S_1 \times \ldots \times S_k| = |S_1| \cdot |S_2| \cdots |S_k| = \prod_{i=1}^{k} |S_i|$$

If all $S_i = S$ (the same set) and $|S| = m$ then $|S^k| = m^k$

NB

This counts the number of sequences where the first item is from S_1, the second is from S_2, and so on.

Example

Let $\Sigma = \{a, b, c, d, e, f, g\}$.
How many 5-letter words?

$$|\Sigma^5| = |\Sigma|^5 = 7^5 = 16,807$$

How many with no letter repeated?

Basic Counting Rules (2)

Product rule: Sequences of selections

Question

How can we count sequences when the underlying set changes?

Answer

- *Define an order on the whole underlying set*
- *Select from* $[1, n]$*, where* n *is the size of the "remaining" set, and a selection of* i *represents choosing the* i*-th element in that set*

Basic Counting Rules (2)

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Example

Let $\Sigma = \{a, b, c, d, e, f, g\}$.
How many 5-letter words with no letter repeated?

$$\prod^{4}(|\Sigma| - i) = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 2,520$$

Basic Counting Rules (2)

Product rule: Sequences with restrictions/duplications

Question

- *How can we count sequences when we have constraints in the underlying order?*
- *How can we count sequences when we have duplicates?*

Example

Let $\Sigma = \{a, b, c, d, e\}$.

- How many 5-letter words with no letter repeated and a before b before c?
- How many 5-letter words can be made from a, a, a, d, e?

Basic Counting Rules (2)

Product rule: Sequences with restrictions/duplications

Question

- *How can we count sequences when we have constraints in the underlying order?*
- *How can we count sequences when we have duplicates?*

Example

Let $\Sigma = \{a, b, c, d, e\}$.

- How many 5-letter words with no letter repeated and a before b before c?
- How many 5-letter words can be made from a, a, a, d, e?

NB

The answer will be the same.

Basic Counting Rules (2)

Product rule: Sequences with restrictions/duplications

- S_1 = sequences with constraints,
- S_2 = ways to define constraints,
- S = sequences without constraints

$$S = S_1 \times S_2,$$

so

$$|S_1| = |S|/|S_2|$$

Alternatively, $\frac{1}{|S_2|}$ of the $|S|$ unconstrained sequences meet the constraint.

Basic Counting Rules (2)

Example

Let $\Sigma = \{a, b, c, d, e\}$.
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Let $\Sigma = \{a, b, c, d, e\}$.
How many 5-letter words with no letter repeated and a before b before c?

Let $\Sigma' = \{a, b, c\}$.

$$\begin{aligned} S &= \prod_{i=0}^{4}(|\Sigma| - i) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120 \\ S_2 &= \prod_{i=0}^{2}(|\Sigma'| - i) = 3 \cdot 2 \cdot 1 = 6 \\ \text{So } S_1 &= 120/6 = 20 \end{aligned}$$

Exercises

Exercises

S, T finite. How many functions $S \longrightarrow T$ are there?

$\boxed{\text{RW: 5.1.19}}$ Consider a *complete* graph on n vertices.

(a) No. of paths of length 3

(b) paths of length 3 with all vertices distinct

(c) paths of length 3 with all edges distinct

Exercises

Exercises

S, T finite. How many functions $S \longrightarrow T$ are there?

$$|T|^{|S|}$$

$\boxed{\text{RW: 5.1.19}}$ Consider a *complete* graph on n vertices.

(a) No. of paths of length 3
Take any vertex to start, then every next vertex different from the preceding one. Hence $n \cdot (n-1)^3$

(b) paths of length 3 with all vertices distinct
$n(n-1)(n-2)(n-3)$

(c) paths of length 3 with all edges distinct
$n(n-1)(n-2)^2$

Exercises

Exercises

$\boxed{\text{RW: 5.3.1}}$ 200 people. 150 swim or jog, 85 swim and 60 do both. How many jog?

$\boxed{\text{RW: 5.6.38}}$ (Supp) There are 100 problems, 75 of which are 'easy' and 40 'important'.
What's the smallest number of easy *and* important problems?

Exercises

Exercises

$\boxed{\text{RW: 5.3.1}}$ 200 people. 150 swim or jog, 85 swim and 60 do both. How many jog?

S – (set of) people who swim, J – people who jog
$|S \cup J| = |S| + |J| - |S \cap J|$; thus $150 = 85 + |J| - 60$ hence $|J| = 125$; answer *does not* depend on the number of people overall

$\boxed{\text{RW: 5.6.38}}$ (Supp) There are 100 problems, 75 of which are 'easy' and 40 'important'.
What's the smallest number of easy *and* important problems?

$$|E \cap I| = |E| + |I| - |E \cup I| = 75 + 40 - |E \cup I| \geq 75 + 40 - 100 = 15$$

Exercise

Exercise

$\boxed{\text{RW: 5.3.2}}$ $S = [100 \ldots 999]$, thus $|S| = 900$.

(a) How many numbers have at least one digit that is a 3 or 7?

(b) How many numbers have a 3 *and* a 7?

Exercise

Exercise

$\boxed{\text{RW: 5.3.2}}$ $S = [100 \ldots 999]$, thus $|S| = 900$.

(a) How many numbers have at least one digit that is a 3 or 7?
$A_3 = \{\text{at least one '3'}\}$ $\qquad$ $A_7 = \{\text{at least one '7'}\}$

$(A_3 \cup A_7)^c = \{\ n \in [100, 999] : n \text{ digits } \in \{0, 1, 2, 4, 5, 6, 8, 9\}\ \}$

7 choices for the first digit and 8 choices for the later digits

$$|(A_3 \cup A_7)^c| = |\{1, 2, 4, 5, 6, 8, 9\}| \cdot |\{0, 1, 2, 4, 5, 6, 8, 9\}|^2$$

Therefore $|A_3 \cup A_7| = 900 - 448 = 452$

(b) How many numbers have a 3 *and* a 7?
$|A_3 \cap A_7| = |A_3| + |A_7| - |A_3 \cup A_7| =$
$(900 - 8 \cdot 9 \cdot 9) + (900 - 8 \cdot 9 \cdot 9) - 452 = 2 \cdot 252 - 452 = 52$

Corollaries

- If $|S \cup T| = |S| + |T|$ then S and T are disjoint
- If $|\bigcup_{i=1}^{n} S_i| = \sum_{i=1}^{n} |S_i|$ then S_i are pairwise disjoint
- If $|T \setminus S| = |T| - |S|$ then $S \subseteq T$

These properties can serve to identify cases when sets are disjoint (resp. one is contained in the other).

Proof.

$|S| + |T| = |S \cup T|$ means $|S \cap T| = |S| + |T| - |S \cup T| = 0$

$|T \setminus S| = |T| - |S|$ means $|S \cap T| = |S|$ means $S \subseteq T$ $\square$

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- Basic counting rules
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Combinatorial Objects: How Many?

permutations

Ordering of all objects from a set S; equivalently: Selecting all objects while *recognising* the order of selection.

The number of permutations of n elements is

$$n! = n \cdot (n-1) \cdots 1, \quad 0! = 1! = 1$$

r-permutations (sequences without repetition)

Selecting any r objects from a set S of size n without repetition while *recognising* the order of selection.

Their number is

$$\Pi(n, r) = n \cdot (n-1) \cdots (n-r+1) = \frac{n!}{(n-r)!}$$

Permutations with duplicates

Example

How many anagrams of ASSESS?

Permutations with duplicates

Example

How many anagrams of ASSESS?
Label S's: $AS_1S_2ES_3S_4$: $6!$
In each anagram we can label the S's in $4!$ ways.
Suppose there are m anagrams. So $m.4! = 6!$, i.e. $m = \frac{6!}{4!}$

Permutations with duplicates

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Example

Number of anagrams of MISSISSIPPI?

Permutations with duplicates

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Example

Number of anagrams of MISSISSIPPI? $\frac{11!}{3!4!2!}$

r-**selections** (or: r-**combinations**)
Collecting any r distinct objects without repetition;
equivalently: selecting r objects from a set S of size n and *not* recognising the order of selection.
Their number is

$$\binom{n}{r} = \frac{\Pi(n,r)}{r!} = \frac{n!}{(n-r)!r!} = \frac{n\cdot(n-1)\cdots(n-r+1)}{1\cdot 2\cdots r}$$

NB

These numbers are usually called binomial coefficients due to

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + b^n = \sum_{i=0}^{n}\binom{n}{i}a^{n-i}b^i$$

Also defined for any $\alpha \in \mathbb{R}$ *as* $\binom{\alpha}{r} = \frac{\alpha(\alpha-1)\cdots(\alpha-r+1)}{r!}$

Simple Counting Problems

Example

$\boxed{\text{RW: 5.1.2}}$ Give an example of a counting problem whose answer is

(a) $\Pi(26, 10)$

(b) $\binom{26}{10}$

Simple Counting Problems

Example

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Draw 10 cards from a half deck (eg. black cards only)
(a) the cards are recorded in the order of appearance
(b) only the complete draw is recorded

Examples

- Number of edges in a complete graph K_n
- Number of diagonals in a convex polygon
- Number of poker hands
- Decisions in games, lotteries etc.

Exercises

Exercises

$\boxed{\text{RW: 5.1.6}}$ From a group of 12 men and 16 women, how many committees can be chosen consisting of

(a) 7 members?

(b) 3 men and 4 women?

(c) 7 women or 7 men?

$\boxed{\text{RW: 5.1.7}}$ As above, but any 4 people (male or female) out of 9 and two, Alice and Bob, unwilling to serve on the same committee.

Exercises

Exercises

$\boxed{\text{RW: 5.1.6}}$ From a group of 12 men and 16 women, how many committees can be chosen consisting of

(a) 7 members? $\binom{12+16}{7}$

(b) 3 men and 4 women? $\binom{12}{3}\binom{16}{4}$

(c) 7 women or 7 men? $\binom{12}{7} + \binom{16}{7}$

$\boxed{\text{RW: 5.1.7}}$ As above, but any 4 people (male or female) out of 9 and two, Alice and Bob, unwilling to serve on the same committee.

$\{\text{all committees}\} - \{\text{committees with both } A \text{ and } B\}$
$= \binom{9}{4} - \binom{7}{2} = 126 - 21 = 105$

equivalently, $\{\text{A in, B out}\} + \{\text{A out, B in}\} + \{\text{none in}\}$
$= \binom{7}{3} + \binom{7}{3} + \binom{7}{4} = 35 + 35 + 35 = 105$

Counting Poker Hands

Exercises

$\boxed{\text{RW: 5.1.15}}$ A poker hand consists of 5 cards drawn without replacement from a standard deck of 52 cards

$$\{A, 2\text{-}10, J, Q, K\} \times \{\text{club, spade, heart, diamond}\}$$

(a) Number of "4 of a kind" hands (e.g. 4 Jacks)

(b) Number of non-straight flushes, i.e. all cards of same suit but *not* consecutive (e.g. 8,9,10,J,K)

Counting Poker Hands

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$$\{A, 2\text{-}10, J, Q, K\} \times \{\text{club, spade, heart, diamond}\}$$

(a) Number of "4 of a kind" hands (e.g. 4 Jacks)
$|\text{rank of the 4-of-a-kind}| \cdot |\text{any other card}| = 13 \cdot (52 - 4)$

(b) Number of non-straight flushes, i.e. all cards of same suit but *not* consecutive (e.g. 8,9,10,J,K)
$|\text{all flush}| - |\text{straight flush}|$
$= |\text{suit}| \cdot |\text{5-hand in a given suit}| -$
$\quad |\text{suit}| \cdot |\text{rank of a straight flush in a given suit}|$
$= 4 \cdot \binom{13}{5} - 4 \cdot 10$

Selecting items summary

Selecting k items from a set of n items:

With replacement	Order matters	Examples	Formula
Yes	Yes	Words of length k (sequences of length k)	n^k
No	Yes	k-permutations	$\Pi(n, k)$
No	No	Subsets of size k	$\binom{n}{k}$
Yes	No		

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Yes	No	Multisets of size k	$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}$

"Balls in boxes"

Have n "distinguishable" boxes.
Have k balls which are either:

1. Indistinguishable
2. Distinguishable

How many ways to place balls in boxes with

A. At most one
B. Any number of

balls per box?

NB

Suppose K is a set with $|K| = k$ and N is a set with $|N| = n$:

- *2A counts the number of injective functions from K to N*
- *2B counts the number of functions from K to N*

"Balls in boxes"

Case	Balls	Balls per box	Number
1A	Indist.	At most 1	
1B	Indist.	Any number	
2A	Dist.	At most 1	
2B	Dist.	Any number	

"Balls in boxes"

Case	Balls	Balls per box	Number
1A	Indist.	At most 1	$\binom{n}{k}$
1B	Indist.	Any number	
2A	Dist.	At most 1	
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"Balls in boxes"

Case	Balls	Balls per box	Number
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"Balls in boxes"

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2A	Dist.	At most 1	$\Pi(n, k)$
2B	Dist.	Any number	n^k

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Alternative techniques

What if the current techniques are unwieldy?
Other techniques for obtaining an exact count:

- Find a different approach for counting
- Make use of symmetries
- Make use of recursion
- Write a program (running time?)

Example

Example

How many sequences of 15 coin flips have an even number of heads?

Example

Example

How many sequences of 15 coin flips have an even number of heads?

- Using "balls in boxes": $\binom{15}{0} + \binom{15}{2} + \ldots + \binom{15}{14}$

Example

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- Use symmetry: $\frac{1}{2} \times 2^{15}$

Example

Example

How many sequences of 15 coin flips have an even number of heads?

- Using "balls in boxes": $\binom{15}{0} + \binom{15}{2} + \ldots + \binom{15}{14}$
- Use symmetry: $\frac{1}{2} \times 2^{15}$
- Use recursion: $\text{Even}(n) = \text{Odd}(n-1) + \text{Even}(n-1)$; $\text{Odd}(n) = \text{Even}(n-1) + \text{Odd}(n-1)$

Example

Example

How many sequences of n coin flips contain HH?

Example

Example

How many sequences of n coin flips contain HH?

$$\begin{aligned} C(0) &= 0 \\ C(1) &= 0 \\ C(n) &= C(n-1) + C(n-2) + 2^{n-2} \end{aligned}$$

Example

Example

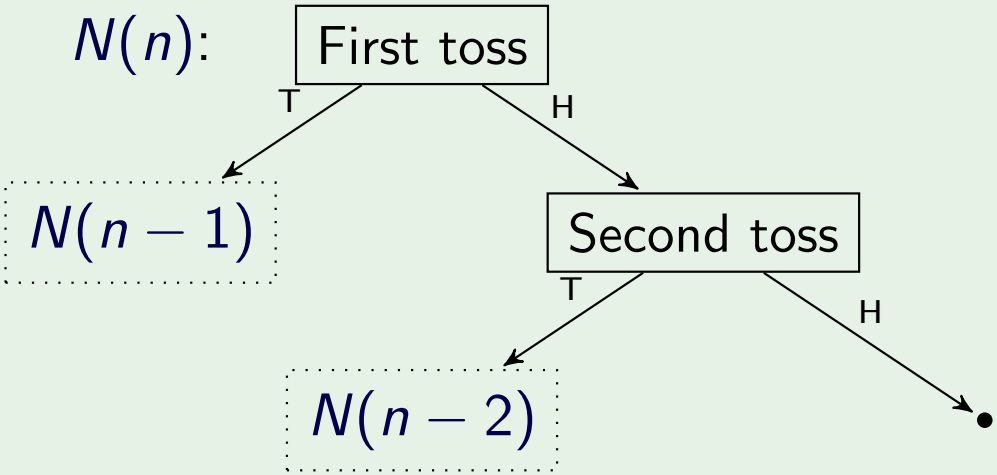
How many sequences of n coin flips do not contain HH?

$$\begin{aligned} N(0) &= 1 \\ N(1) &= 2 \\ N(2) &= 3 \\ N(n) &= N(n-1) + N(n-2) \end{aligned}$$

Example

Example

How many sequences of n coin flips do not contain HH?
We can summarise all possible outcomes in a **recursive tree**



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Difficult Counting Problems

Example (Ramsay numbers)

An example of a *Ramsay number* is $R(3,3) = 6$, meaning that

> *"K_6 is the smallest complete graph such that if all edges are painted using two colours, then there must be at least one monochromatic triangle"*

This serves as the basis of a game called S-I-M (invented by Simmons), where two adversaries connect six dots, respectively using blue and red lines. The objective is to *avoid* closing a triangle of one's own colour. The second player has a winning strategy, but the full analysis requires a computer program.

Using Programs to Count

Two dice, a red die and a black die, are rolled.
(Note: one *die*, two or more *dice*)

Write a program to list all the pairs $\{(R, B) : R > B\}$

Similarly, for three dice, list all triples $R > B > G$

Generally, for n dice, all of which are m-sided ($n \leq m$), list all *decreasing* n-tuples

NB

In order to just find the number of such n-tuples, it is not necessary to list them all. One can write a recurrence relation for these numbers and compute (or try to solve) it.

Approximate Counting

NB

A Count may be a precise value or an **estimate**.

The latter should be asymptotically correct or at least give a good asymptotic bound, whether upper or lower. If S is the base set, $|S| = n$ its size, and we denote by $c(S)$ some collection of objects from S we are interested in, then we seek constants a, b such that

$$a \leq \lim_{n \to \infty} \frac{est(|c(S)|)}{|c(S)|} \leq b$$

In other words $est(|c(S)|) \in \Theta(|c(S)|)$.